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The Reconstruction of a Protein Backbone from C $^{\alpha}$ Coordinates

A Monte Carlo Protein Building method to construct the backbone and C $^{\beta}$ atomic coordinates from known C $^{\alpha}$ coordinates is reported. The method selects values of dihedral angles from ranges established from a statistical analysis of the relationship between dihedral angles of the backbone and C $^{\alpha}$ coordinates for a protein data base. The averaged coordinates from ten backbone models of a protein were used to define a mean structure that was refined by energy minimization using the AMBER force field (GB/SA). By the latter method the average atomic deviation and rmsd of the backbone and C $^{\beta}$ atoms for 24 proteins is between 0.14 and 0.32 Å (average 0.22 Å) and 0.22 and 0.61 Å (average 0.43 Å), respectively. A comparison with other methods is made. © 1997 John Wiley & Sons, Inc.

INTRODUCTION

In recent years the construction of a full-atom representation of a protein from C $^{\alpha}$ coordinates has received considerable attention. Commonly employed procedures¹ make use of "backbone dictionaries" and are referred to as homology building methods. The protein sequence and C $^{\alpha}$ coordinates of a target protein are compared with a data base of crystal structure coordinates for segments of the same sequence and similar topology. The all-atom atomic coordinates of the residues in the analogous x-ray crystal structures are transposed to the target protein. The method has limitations in regions where no similar patterns of sequence and topology exist in the data base, and in the linking regions of segments of known topology.

Several methods that avoid these problems have been developed including those of Purisima and

Scheraga,² Reid and Thornton,³ Correa,⁴ Holm and Sander,⁵ Jones and Thirup,⁶ and Rey and Skolnick.⁷ For example, Correa constructed the protein sequentially from C $^{\alpha}$ coordinates with energy minimization after positioning each backbone and β -carbon atom, but this is expensive in computer time even for medium-sized proteins. More recently, Mathiowetz and Goddard⁸ have reported a Dihedral Probability Grid Monte Carlo method to build protein models from C $^{\alpha}$ coordinates. The dihedral angles of the backbone and the torsional angles of the side chains are randomly selected for each residue from predetermined ranges in a probability matrix. The probabilities were assigned to the protein backbone and side-chain dihedrals according to their distributions in known protein structures. Mandal and Linthicum⁹ have described a modeling algorithm, PRO-GEN, which uses an optimal geometry parameter database established by examining the statistical

Received February 29, 1996; accepted July 9, 1996.

Biopolymers, Vol. 41, 381–389 (1997)

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CCC 0006-3525/97/040381-09

correlation among 23 different intra- and interpeptide geometric parameters relating C^α distances for each amino acid in a library of nineteen proteins from the PDB. The initial model is refined by energy minimization and molecular dynamics techniques while keeping the C^α atoms fixed.

We now report a Monte Carlo Protein Building (MCPB) method to reconstruct the backbone and C^β atomic coordinates of 24 proteins from known C^α coordinates. The first problem addressed concerns the positioning of the atoms of the first peptide. After the first peptide is positioned a Monte Carlo method for selecting values of bond lengths and angles and dihedral angles from predetermined ranges is used along with criteria for their acceptance to reconstruct atomic coordinates of the backbone and β -carbons.

METHOD

Modeling and computations were performed on an IBM RS/6000. The programs *mtranscoil*, *crystal*, *calcsup*, *calcrms*, and *range* were written in FORTRAN 77. Statistical analysis was carried out using the program SAS¹⁰ running on a VAX750/VMS system. The program *pbdmmod* was used to convert PDB structure files to Macromodel format (Macromodel package, Department of Chemistry, Colombia University, New York, NY 10027). Bond lengths, bond angles, and dihedral angles were determined with the program *mtranscoil*. A target structure was generated by the program *crystal* and the detail of this program and method is detailed below. The program *range* was used to determine the boundaries of v_6 and v_{13} in the PDB. Energy minimization was carried out with *Macromodel/Batchmin2K* when the protein contained less than 2000 backbone atoms (i.e., 400 residues) or *Batchmin5K* for larger proteins with up to 5000 atoms, e.g., 1TIM, 1PSG, 1CSE, and 3RP2. The rmsd between a model and the crystal structure was calculated using the program *calcsup*, which superimposes two structures by the method developed by Kabsch.^{11,12} The program was also used for analysis for each type of amino acid and of the different motif regions. The results of rmsd calculations were checked using *PCmodel* software (PCMODEL 4.0 version, Serena Software, Box 3076, Bloomington, IN 47402-3076). The motif of an amino acid in the analysis was as defined in the PDB. Two methods are applied to generate the atomic coordinates of the backbone from C^α coordinates. Our study has focused on the proteins selected because of their diversity and the accuracy with which the crystal structure data is known.

Generation of the First Peptide

The first problem to address in building the protein concerns the positioning of the atoms of the first peptide

Table I The Bond Length and Bond Angles

	Bond Length (Å)		Bond Angle (°)
$C^\alpha-C^\beta$	1.540 ± 0.04	$N-C^\alpha-C^\beta$	111.0 ± 3.5
$C^\alpha-C'$	1.530 ± 0.04	$N-C^\alpha-C'$	110.0 ± 3.0
$C'-O$	1.225 ± 0.03	$C^\alpha-C'-O$	122.5 ± 2.5
$C'-N$	1.340 ± 0.04	$C^\alpha-C'-N$	112.0 ± 5.0
$N-C^\alpha$	1.450 ± 0.04		

and is important since it is from these atoms that further construction occurs. We have chosen to build the protein from the N-amino acid end. The program *crystal* chooses which of the terminal amino acid configurations is appropriate from criteria defined later. We first describe building the protein with the first amino acid in a right-handed α -helical conformation. The carbonyl carbon of the first amino acid from the N-terminus has been defined in the following way. A statistical analysis of the α -helical regions of the 14 proteins (1CRN, 1CTF, 1PPT, 1RDG, 1ROP, 1UTG, 2CCY, 2OVO, 2ZTA, 3FXN, 3CLA, 3WRP, 4PTI, 4TNC)¹³ shows the carbonyl carbon C'_1 of the first amino acid lies close to a plane defined by $C_1^\alpha, C_2^\alpha, C_5^\alpha$ and therefore this atom in the first amino acid is positioned on this plane. Similar analysis of the crystal structures of 14 proteins, but irrespective of motif, showed the angle¹⁴ $C', C_1^\alpha, C_2^\alpha$ to be ca. 21.5° and therefore this value was chosen in defining the first peptide carbonyl carbon, C' with respect to C_1^α and C_2^α . The C'_1, C_1^α bond length of the first amino acid was fixed at the average value of 1.540 Å (Table I).¹⁵ Solution of three equations that relate the atomic coordinate of C'_1 to the C'_1, C_1^α bond length and the angle $C', C_1^\alpha, C_2^\alpha$ as 21.5° and define the coordinates of C'_1 on the plane $C_1^\alpha, C_2^\alpha, C_5^\alpha$ give rise to four possible solutions for the C'_1 coordinates. The solution that is accepted is chosen to meet the requirement that the distance between C'_1 and C_5^α and the distance between C'_1 and C_2^α be less than 6.5 and 2.8 Å respectively thus requiring C'_1 to lie between C_1^α and C_2^α .

After the coordinates of carbonyl carbon C'_1 are defined, the atomic coordinates of N_2 and O_1 are calculated from the carbonyl carbon (C'_1), C_1^α , and C_2^α by using the Scheraga algorithm¹⁶ which transforms the internal coordinates into the Cartesian coordinates. For computing the nitrogen coordinates a C'_1, N_2 bond length and C_1^α, C'_1, N_2 angle is required and these are randomly selected from within the ranges $1.35 \pm 0.05 \text{ Å}$ and $117 \pm 5^\circ$ respectively (Table I). The dihedral angle $N_2, C'_1, C_1^\alpha, C_2^\alpha$ is set to zero as for a planar peptide bond. For the generation of O_1 coordinates the algorithm requires the carbonyl C'_1, O_1 bond length, which is randomly selected within the range $1.23 \pm 0.03 \text{ Å}$ and an angle for C_1^α, C'_1, O_1 selected within the range $122.5 \pm 2.5^\circ$. The dihedral angle $O_1, C'_1, C_1^\alpha, C_2^\alpha$ were chosen at 180° as for a planar peptide bond. After establishing the N_2 and O_1

coordinates, the distance between N_2 and C_2^α is computed. If the distance is in the range of $1.45 \pm 0.04 \text{ \AA}$, consistent with a normal bond length, then the coordinates of N_2 and O_1 are accepted and if outside this range the coordinates of both these atoms are regenerated until the distance is within the defined range.

To establish the coordinates of N_1 the same algorithm is used. A value for the N_1, C_1^α bond length is randomly selected from within the range $1.45 \pm 0.04 \text{ \AA}$ and a N_1, C_1^α, C_1' bond angle is similarly selected from within the range of $107^\circ \pm 5^\circ$. A value for the $N_1, C_1^\alpha, C_1', N_2(\psi_1)$ dihedral angle is selected from the range $-40^\circ \pm 20^\circ$ typical for right-handed α -helical proteins.^{17,18} The C^β is defined by the same algorithm requiring the distance C_1^β, C_1^α , randomly selected from the range $1.54 \pm 0.04 \text{ \AA}$ and an angle for $C_1^\beta, C_1^\alpha, C_1'$ from the range $107^\circ \pm 5^\circ$. The dihedral angle $C_1^\beta, C_1^\alpha, C_1', N_2$ is $\psi_1 + 120^\circ$, where ψ_1 is chosen from the range $-40^\circ \pm 20^\circ$.

The same method is used for generating the first amino acid in a β -sheet, a β -turn, or a random-coil motif except that the assumption is made in all these motifs that the carbonyl carbon C_1' lies on the $C_1^\alpha, C_2^\alpha, C_3^\alpha$ plane rather than on the $C_1^\alpha, C_2^\alpha, C_5^\alpha$ plane. An analysis of the nonhelical regions of the 14 proteins showed this to be close to reality. An analysis of nonhelical structure motifs in 14 proteins show the C_1', O_1, N_2 plane makes a dihedral angle with the $C_1^\alpha, C_2^\alpha, C_3^\alpha$ plane of $20^\circ \pm 50^\circ$ and therefore the assumption has limitations. To establish the coordinates of N_1 and C_1^β for these motif regions ψ_1 is chosen from the range $150^\circ \pm 30^\circ$ for the β -sheet motif. The β -turn and random-coil motifs are not differentiated and ψ_1 is chosen from the range $80^\circ \pm 50^\circ$.^{17,18}

Generation of the Backbone Atoms of the Second and Subsequent Amino Acids

The second amino acid from the N-amino end is built to the first amino acid (whether as an α -helical configuration or a nonhelical configuration) in the following way. The method involves the use of data bases and a Monte Carlo method of selecting values from predetermined ranges in values along with criteria for their acceptance. The predetermined ranges of the variables used above for building both the first peptide and the second to the last amino acid of the chain were established from an analysis of the x-ray structures of fourteen proteins.¹⁹ The proteins were chosen because the x-ray crystal structure coordinates were known with an accuracy of better than $2.5 \text{ \AA}$.^{20,21}

Since the coordinates of the nitrogen and C^α of the second amino acid are already defined by the first peptide, the dihedral angle ϕ_2 will define the positions of C^β and the carbonyl carbon of the second amino acid along with appropriate bond lengths and bond angles. In the same way the dihedral angle ψ_2 will define the positions of the nitrogen of the third amino acid and the carbonyl oxygen of the second amino acid. A data base for dihedral angles was established from a statistical analysis of the relation-

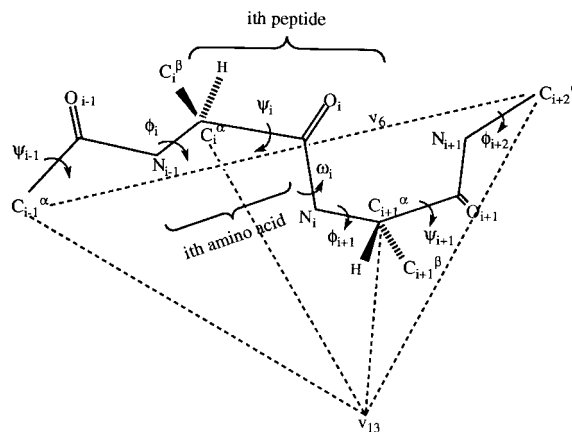


FIGURE 1 Definition of v_6 , v_{13} , ϕ , and ψ .

ship between dihedral angles of the backbone and C^α coordinates for a sample of 14 proteins using SAS. We were interested to see if there is a relationship between values of v_6 and v_{13} and ϕ and ψ that could provide a satisfactory data base from which dihedral angles could be selected in modeling the backbone. For the 14 protein selection we have investigated the distribution of ϕ_i and ψ_i for the i th amino acid with v_6 and v_{13} for that residue (see Figure 1). We first divided the values of v_6 and v_{13} into four arbitrary ranges: v_6 with boundaries (1) 4.0–6.0 $\text{\AA}$, (2) 6.0–8.0 $\text{\AA}$, (3) 8.0–9.5 $\text{\AA}$, (4) 9.5–13.0 $\text{\AA}$, respectively; and v_{13} into quadrants (1) 180° to 90° , (2) 90° to 0° , (3) 0° to -90° , (4) -90° to -180° , respectively. For example, for the i th amino acid residue, if the distance v_6 (between C_{i-1}^α to C_{i+2}^α) is in the range 4.0–6.0 $\text{\AA}$, then v_6 is defined to be in range (1) and if the torsion angle v_{13} ($C_{i-1}^\alpha, C_i^\alpha, C_{i+1}^\alpha, C_{i+2}^\alpha$) is between 90° to 0° , v_{13} is defined to be in range (2). The C_i^α atom is then recorded as in range (1) for v_6 and range (2) of v_{13} . For the sake of simplification this is recorded as REGION 12—the first number of a REGION representing a range of v_6 and the second the v_{13} value. There are 16 such REGIONS from 11 through 44.

The mean values and standard deviations of the dihedral angles that have values of v_6 and v_{13} corresponding to the ranges of these parameters defined in the REGIONS 11–44 above were determined for the 14 proteins in the data base. The arbitrary boundaries of v_6 and v_{13} defined above for REGIONS 11–44 were adjusted so as to give a minimum overall average standard deviation in the values of ϕ and ψ . The boundaries that gave rise to the lowest overall standard deviation is defined in Table II. No dihedral angles were observed for the proteins in the REGIONS 11, 14, and 42. The values of ϕ and ψ in these REGIONS as reported in the table are taken from analysis of the twenty four protein sample along with literature reports of dihedral angles in β -sheets.^{17,18}

The boundaries of the REGIONS 11–44 from which ϕ and ψ values are taken were further optimized by repetitive small empirical changes to the centre of the v_6 and

Table II Mean and Standard Deviations in the Values of ϕ and ψ in REGIONS 11–44*

Region	− ϕ			+ ϕ			− ψ			+ ψ		
	<i>u</i>	σ	<i>P</i> ^c	<i>u</i>	σ	<i>P</i> ^c	<i>u</i>	σ	<i>P</i> ^c	<i>u</i>	σ	<i>P</i> ^c
12	−63.6	7.8	99.6	66.1	32.1	0.4	−40.1	10.3	99.3	91.6	52.6	0.7
13	−62.3	†0.0 ^b	33.3	35.6	25.7	66.7	0.00	0.00	0.00	46.1	10.3	100
21	−104.2	13.3	68.2	73.0	10.4	31.8	−10.1	0.00	4.5	26.9	16.1	95.5
22	−82.7	21.9	92.4	101.6	28.0	7.6	−37.7	29.7	77.2	81.6	64.9	22.8
23	−84.6	16.3	91.7	93.2	13.2	8.3	−18.8	14.8	38.9	104.9	70.7	61.1
24	−82.9	17.7	100.0	0.00	0.00	0.0	−17.4	11.9	66.7	4.7	1.58	33.3
31	−98.8	19.9	58.0	76.9	18.5	42.0	−18.3	20.4	12.0	39.2	34.6	88.0
32	−112.3	17.8	57.1	91.5	5.50	42.9	−8.8	2.1	71.4	5.5	1.03	28.6
33	−87.6	29.5	90.0	106.0	32.2	10.0	−66.9	68.5	10.0	154.4	13.8	90.0
34	−87.6	23.9	97.7	168.9	0.00	2.3	−19.5	11.5	22.7	118.8	37.4	77.3
41	−109.3	25.4	80.5	88.0	17.9	19.5	−8.3	7.2	7.3	100.2	49.0	92.7
43	−112.9	32.2	96.9	167.5	0.00	3.1	−168.5	6.7	6.1	156.9	11.9	93.9
44	−111.1	27.3	98.4	51.8	10.3	1.6	−79.1	66.6	2.5	137.7	22.9	97.5
Av σ ^d		21.1			19.4			22.7			29.8	

^a Definitions of REGIONS of v_6 and v_{13} that result in minimum overall average standard deviation in values of ϕ and ψ are as follows: v_6 (1) 4.0–5.84 Å, (2) 5.84–8.09 Å, (3) 8.09–9.10 Å, (4) 9.10–13.0 Å; v_{13} (1) 180° to 99.62°, (2) 99.62° to −8.97°, (3) −8.97° to −118.93°, (4) −118.93° to −180°.

^b Only a single value in the data base for the 14 proteins.

^c Probability that a dihedral angle of a particular amino acid from the 14 proteins will occur in this range.

^d The average standard deviations for all regions is 23.25°.

v_{13} ranges. This was effected using the program crystal. The total number of variations examined was greater than 4000. The ranges that allowed for prediction of a backbone structure of the twenty four proteins by the MCPB method with minimum rmsd's are shown in Table IV. Values of ϕ and ψ for an amino acid residue reflect the twist of the protein and can be both positive, both negative or one positive and the other negative. The table shows the preference for a positive or negative value of the dihedral angle.

In building the coordinates of the backbone values of v_6 and v_{13} for each amino acid residue are established from the C^α coordinates with the subroutine *cavara* in crystal. The REGIONS 11–44 for each C_i^α is then established on the basis of the optimum ranges as defined in Table III. This also includes the first amino acid, the coordinates of which are then established using the method described above.

The second amino acid from the N-amino end is built to the first amino acid, whether as an α -helical configuration or a nonhelical configuration. A value for each of the dihedral angles ϕ and ψ for the amino acid is randomly selected according to the assigned REGION, taking into account the more probable sign combination of the dihedral angles. Appropriate bond lengths and angles are selected by a random number method from the values defined in Table I. If the resulting coordinates meet the criteria defined earlier, they are accepted.

In building a residue, i.e., the second or the i th residue, it is possible to choose +ve or −ve values for ϕ and

ψ . Of the four combinations (++, --, +−, −+) the probability is not the same. In reconstructing each amino acid of the protein, the choice of sign of ϕ and ψ can be (a) the more probable sign of ϕ and ψ or (b) the more probable sign of ϕ and less probable of ψ (c) less probable sign of ϕ and more probable of ψ (d) less probable sign of ϕ and less probable of ψ .

Two criteria for acceptance have been found effective and depend on values of the distance d_i and torsional angle ω_i (Figure 2): (1) the bond length for the i th peptide unit d_i , between N_i and C_{i+1}^α of the next peptide must be in the range $1.40 \text{ Å} \leq d_i \leq 1.50 \text{ Å}$ and (2) the torsion angle ω_i for a *trans* peptide $|\omega_i| \geq 170^\circ$ and for a *cis* peptide $|\omega_i| \leq 10^\circ$. The distance between C_i^α and C_{i+1}^α and the absolute value of ω allow *cis* and *trans* peptide configurations to be recognized. Analysis of the 14 proteins showed the distance between C_i^α and C_{i+1}^α for a *trans* peptide is in the range $3.80 \pm 0.35 \text{ Å}$ and for a *cis* peptide is in the range $2.8 \pm 0.65 \text{ Å}$. If the values of d_i and ω_i are within these criteria, the coordinates are accepted and the atomic coordinates of the next peptide generated. If the values are outside these specified ranges then the coordinates of atoms in this peptide unit are rejected and the random seed generator is used to select alternate values and the process repeated.

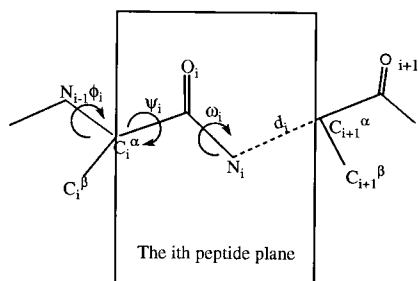
The program crystal is written such that, in the building of the protein, combination (a) is chosen first, and if the criteria for accepting the coordinates so produced is not met within 500 cycles then combination (b) is chosen,

Table III Range in Negative and Positive Values of ϕ and ψ for Each Region

REGION	ϕ		ψ	
	Range 1 ^a	Range 2 ^b	Range 1 ^c	Range 2 ^d
11	-45 ± 25	45 ± 25	-5 ± 40	5 ± 40
12	-62 ± 24	35 ± 50	-40 ± 20	90 ± 60
13	-50 ± 25	36 ± 50	47 ± 30	-90 ± 60
14	45 ± 35	-45 ± 35	30 ± 30	-30 ± 30
21	-105 ± 40	75 ± 35	30 ± 35	-10 ± 35
22	-80 ± 40	100 ± 50	-40 ± 50	80 ± 60
23	-85 ± 40	95 ± 30	105 ± 65	-20 ± 30
24	-80 ± 40	80 ± 40	-20 ± 30	10 ± 25
31	-100 ± 40	77 ± 45	40 ± 60	-20 ± 40
32	-110 ± 40	90 ± 25	-10 ± 20	10 ± 40
33	-90 ± 50	105 ± 50	155 ± 25	-70 ± 60
34	-90 ± 50	170 ± 10	120 ± 40	-20 ± 33
41	-110 ± 50	90 ± 35	100 ± 55	-10 ± 20
42	-130 ± 40	125 ± 45	155 ± 25	150 ± 30
43	-110 ± 40	150 ± 35	155 ± 30	-170 ± 10
44	-110 ± 50	50 ± 30	140 ± 50	-80 ± 60

^a More probable sign of ϕ .^b Less probable sign of ϕ .^c More probable sign of ψ .^d Less probable sign of ψ .

and so on. If suitable values of d_i or ω_i are not determined within 1000 cycles, the next most probable combination of sign of ϕ and ψ are used to generate the coordinates, and so on. If suitable values are not found within 1500 cycles then the final combination is used, and if suitable values are not found within 2000 cycles then the program returns to the most probable sign combination but the criteria are relaxed by 0.01 Å for d_i or 1° for ω_i , e.g., $1.39 \text{ Å} \leq d_i \leq 1.50 \text{ Å}$ and $|\omega_i| \geq 169^\circ$ or $|\omega_i| \leq 11^\circ$ until $1.35 \text{ Å} \geq d_i$ or $d_i \geq 1.55 \text{ Å}$ and $|\omega_i| \leq 140^\circ$ or $|\omega_i| \geq 40^\circ$ for the *trans* and *cis* peptide, respectively, are met. After a set of coordinates are accepted, the acceptance ranges for d_i and ω_i are reset to their initial default values and most probable sign combination before generating the next peptide unit. The process is repeated for each amino acid residue of the backbone of the protein.

**FIGURE 2** Definition of d_i and ω_i .

Since the C^α parameters cannot be defined at the carbonyl terminus of the last residue, the terminal N atom is capped with a methyl group. The procedure described is encapsulated in the program crystal and the method we describe as Monte Carlo Protein Building (MCPB).

Mean Coordinates of the Backbone Model

Levitt was the first to average coordinates from several model structures of a protein.²² We have similarly shown that the rmsd of an average or mean structure of several model structures with the x-ray structure was smaller than for any individual model. As a result we have examined this methodology in some detail. The bond lengths and bond angles in the averaged model structure are checked for distortions and to see if they are in the ranges defined in Table I. If they are outside these ranges, crystal is programmed to recognize the structure as distorted. If a structure is distorted energy minimization can bring the bond lengths and angles to sensible values. To date we have not found any averaged structure of protein backbone to be distorted. An empirical study was made to determine if there was an optimum number of independent models that should be averaged to give a mean structure with the lowest energy rmsd. It was found that little benefit is gained in averaging more than ten structures.

Energy Minimization of the Backbone Model

An average structure (of ten structures) without hydrogen atoms for each of the 24 proteins was minimized with Macromodel/Batchmin using the OPLS/AMBER (non-hydrogen) force field²³ and GB/SA water model.²⁴ The program uses default values of the cutoff distances of nonbonded and electrostatic interactions. The C α atoms of the model structure are restrained to their initial positions as defined by the x-ray coordinates.

We examined the effect of removing the constraints on the C α atomic positions during minimization for the mean structures. When the constraints on the C α positions are released the coordinates of the C α positions move and the rmsd necessarily increases, by inclusion of the now variable C α positions, to ca. 1.1 Å. The results show that after minimization without constraint on the C α positions the rmsd is increased. For the subsequent studies it was considered pertinent to effect minimization with the C α constraints in place.

RESULTS AND DISCUSSION

The MCPB method has been applied to the twenty four proteins.¹³ An advantage of the method is that it is capable of generating the structure of proteins that contain subunits or peptide chains that are not linked. The average atomic deviation and the rmsd of the backbone and C β atoms of the mean modeled structures and the x-ray structures for the twenty-four proteins are listed in the Table IV.

Energy minimization with the C α atoms fixed reduces the rmsd by ca. 0.1 Å. For the backbone atoms the average atomic deviation is 0.35 Å and rmsd 0.53 Å, and these values reduce to 0.22 and 0.43 Å, respectively, after energy minimization. The proteins with the higher content of helical structure, e.g., 1AMT, 3WRP, and 4TNC and the coiled-coil proteins, 2ZTA and 1ROP, give the lowest rmsd values. The backbone atoms, nitrogen, carbonyl carbon, oxygen including C β have rmsd's of 0.29, 0.26, 0.77, and 0.36 Å, respectively. Oxygen shows a greater variation than C β .

The mean model structures of each of the 24 proteins were examined after energy minimization by motif and the rmsd for the amino acids in each motif region calculated. For this analysis regions of α -helix, β -sheet, and β -turn are as defined in the PDB file, and the remainder of the structure is assumed to be random coil. In general, the rmsd of the secondary structure is lowered by energy minimization with the C α coordinates fixed. The average rmsd of α -helical region, β -sheet region, β -turn region, and random coil region are 0.31, 0.42, 0.58,

and 0.57 Å, respectively. The α -helical region gave the lowest deviation. The rmsd of β -sheet regions is lower than the β -turn and random coil regions, but higher than the average rmsd.

The rmsd of each of the 20 amino acids in the averaged models of the 24 proteins was also examined. The number of amino acids included in the analysis is 3536. (There are 27 undefined amino acid residues in the protein 1AMT that have no definition in its PDB file.) After energy minimization of the averaged structures the overall average rmsd for all the amino acids was 0.47 Å. Glutamine (E) showed the minimum rmsd (0.37 Å) and glycine (G) the maximum (0.66 Å). The total number of proline residues in the 24 proteins is 151 (141 *trans* and 10 *cis*) and the rmsd after energy minimization was of 0.53 Å (*trans* 0.53 and *cis* 0.61 Å).

Energy minimization of the structure increases the percentage of the values of ϕ and ψ within a $\pm 30^\circ$ window of the values in the x-ray structure for the proteins. The overall percentage of values of ϕ and ψ within a $\pm 30^\circ$ window for the 24 proteins is 87 and 86%, respectively. Proteins with a high helical content, e.g., 2ZTA, 1ROP, 1UTG, 3WRP, and 4TNC, give a high percentage > 90% of dihedral angles within the $\pm 30^\circ$ window. For the proteins with a high content of β -turn and random-coil segments, e.g., 1CRN, 1PCY, 1PSG, 1TIM, 2LYM, 2LYZ, 4PTI, 4INS, and 3RP2, the accuracy of prediction within the window is below 85%.

The overall percentages of dihedral angles ϕ and ψ for the 20 amino acids within a $\pm 30^\circ$ window of the values found from the x-ray structure are 84 and 83%, respectively. For most amino acids 80% of the values of ϕ and ψ fall within a $\pm 30^\circ$ window. The greatest deviation is observed for the dihedral angles of glycine, with no side-chain constraints, where 75% of the values of ϕ and 72% of the values of ψ fall within the $\pm 30^\circ$ window. For proline, 83% of the values of ϕ and 83% of the values of ψ , respectively, fall within the $\pm 30^\circ$ window. The values for *cis* and *trans* proline are comparable. Proline therefore parallels the other amino acids. In all cases energy minimization has the effect of increasing the percentage of values of ϕ and ψ that fall within the $\pm 30^\circ$ window by ca. 10%.

Comparison with Previous Methods

In comparing the ability of the MCPB method to model protein coordinates with these other methods, three criteria are considered; the rmsd of the backbone and C β atoms with the x-ray structure, the information required for the data base to build the

Table IV The Average Atomic Deviation and RMSD of the Backbone Models (Averaged)

Protein	N^a	Ave Dev ^b	RMSD ^c	Ave Dev ^d	RMSD ^e
1AMT	60	0.26	0.41	0.18	0.36
1CRN	46	0.39	0.63	0.23	0.47
1CSE	337	0.39	0.59	0.27	0.55
1CTF	68	0.38	0.59	0.24	0.48
1PCY	99	0.44	0.64	0.26	0.53
1PPT	36	0.40	0.59	0.18	0.39
1PSG	365	0.39	0.59	0.29	0.55
1RDG	52	0.39	0.57	0.23	0.43
1ROP	56	0.23	0.36	0.14	0.30
1TIM	494	0.37	0.56	0.29	0.52
1UTG	70	0.27	0.38	0.14	0.24
2CCY	254	0.29	0.48	0.19	0.41
2LYM	129	0.42	0.67	0.31	0.61
2LYZ	129	0.41	0.67	0.32	0.61
2MHR	117	0.33	0.57	0.19	0.48
2OVO	56	0.36	0.50	0.19	0.33
2ZTA	62	0.21	0.28	0.14	0.22
3FXN	138	0.34	0.52	0.23	0.43
3CLA	125	0.32	0.47	0.18	0.33
3RP2	448	0.39	0.57	0.25	0.49
3WRP	101	0.23	0.37	0.15	0.29
4INS	102	0.39	0.64	0.23	0.50
4PTI	58	0.41	0.60	0.29	0.57
4TNC	160	0.28	0.40	0.17	0.28
Overall		0.35	0.53	0.22	0.43

^a The number of residues in the protein.^b Average deviation (Å) before energy minimization.^c The rmsd (Å) before energy minimization of the model structure.^d Average deviation (Å) after energy minimization.^e rmsd (Å) after energy minimization.

backbone, and the computational requirements. The method pioneered by Jones and Thirup⁶ uses a data base or dictionary of structural templates from known structures. Reid and Thornton³ applied this method to the generation of the main-chain atomic coordinates of 3FXN and obtained a rmsd for the main chain 0.51 Å. Claessens et al.²⁵ was the first to use an automatic search segment data base for building the backbone atoms of three proteins—triose phosphate isomerase, citrate synthase and carboxyl peptidase—and obtained an average rmsd of 0.61 Å, somewhat poorer than the average rmsd (0.43 Å) obtained in the present studies.

A method called “segment match modelling,” which requires the amino acid sequence and a data base of x-ray structures, has been developed by Levitt.²² The method also positions C^β atoms. An average rmsd of the main-chain atoms excluding C^β atoms of 0.42 Å for 8 proteins was obtained. The method gave a satisfactory result for 1CTF and

4PTI, but the results for 1CRN and 3FXN are not as good. Recently, Mandal has developed PRO-GEN⁹ that uses an optimal geometry parameter data base established from a statistical analysis employing 23 different intrapeptide and interpeptide geometric parameters relating to C^α coordinates for 19 proteins for the positioning of atoms for each amino acid. The rmsd for the backbone atoms and C^β ranged from 0.29 to 0.76 Å (average 0.53 Å) and with average atomic deviations between 0.14 and 0.44 Å (average 0.28 Å).

Correa⁴ used molecular dynamics methodology to build protein structures from the C^α positions in a method that does not use any data base of known protein structural information. The rmsd of backbone atoms for three proteins (α -lytic protease, troponin C, and flavodoxin) using this method are 0.19, 0.41, and 0.49 Å, respectively; however the method requires considerable computer time. Recently, Rey and Skolnick developed a purely analyt-

Table V The RMSD After Energy Minimization for Eight Different Modeling Methods^a

Protein	RMSD (Å)							
	M1	M2	M3	M4	M5	M6	M7	M8
3FXN	0.43	0.51	0.48	0.49	0.44	0.49	0.46	0.51
1CTF	0.48	—	—	—	0.29	—	—	—
1CRN	0.47	—	—	—	0.56	—	—	0.43
1TIM	0.52	—	0.59	—	—	—	0.53	—
4TNC	0.28	—	0.41	—	—	—	0.36	—
4PTI	0.57	—	—	0.43	0.51	—	0.45	—

^a Method (M) 1: MCPB, this work. M2: Reid and Thornton.³ M3: Holm and Sander.⁵ M4: Rey and Skolnick.⁷ M5: Levitt.²² M6: Correa.⁴ M7: Mandal and Linthicum.⁹ M8: Mathiowetz and Goddard.⁸

ical method⁷ for generating a protein backbone from C^α coordinates and used the method to build the backbone structures of 6 proteins. The average rmsd of the backbone atoms was 0.70 Å before energy minimization and 0.49 Å after energy minimization. A comparison of the results from different methods is shown in Table V.

The amino acid sequence of a protein is not required in the generation of the model of backbone atoms including C^β atoms in our MCPB method; however, the sequence is required for positioning of the side chains. This has its value. For example, the protein 1AMT has 61 amino acid residues and 27 of them have no definition in the PDB file. The MCPB method can generate the backbone atomic coordinates, including C^β atoms, from the C^α coordinates and give a model with a reasonable rmsd. This method at this stage does require a full set of C^α coordinates. The method also has an advantage of being computer inexpensive. For the generation of an average model using the program crystal approximately 0.5 s of computer time is required per residue. Energy minimization requires approximately 9 s of computer time per residue.

CONCLUSION

The MCPB method reported herein allows for the construction of a protein backbone from C^α coordinates. The method requires the C^α coordinates to be known but the sequence is not required. The method gives the backbone structures whose coordinates deviate from the x-ray coordinates by an average of 0.52 Å before energy minimization and 0.43 Å after energy minimization with the OPLS/AMBER force field and GB/SA solvent method. The method compares favorably with other methods

and is cost effective even with energy minimization. The method is efficient and robust for the modeling of protein backbones and has the potential when integrated into and used in conjunction with traditional x-ray techniques to speed up structure solution. A simulated annealing procedure to generate side-chain conformation will be reported in the subsequent paper.

This work was funded by the New Zealand Foundation for Research, Science and Technology. We acknowledge grants from the New Zealand Lotteries Board.

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